What do we mean by uncertainty estimation?

What is uncertainty?

Uncertainty as level of confidence

- Any concrete concept can have (and does always have) some level of uncertainty.
  - In machine learning, there can be many points of uncertainty:
    - uncertainty of evaluation being correct,
    - uncertainty on the model being fit,
    - uncertainty on data being correctly labelled,
    - uncertainty on the optimization finding the right solution,
    - etc,
- Confidence in prediction: i.e., Predictive Uncertainty

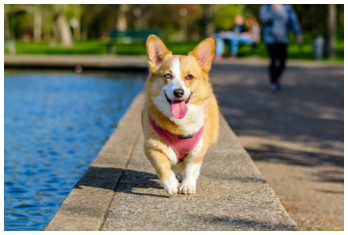


What do we mean by *predictive* uncertainty estimation?

- We want the model to be able to say, "I don't know!", or, this is my prediction and "I have no doubt!".
- Or essentially how confident I am in my prediction

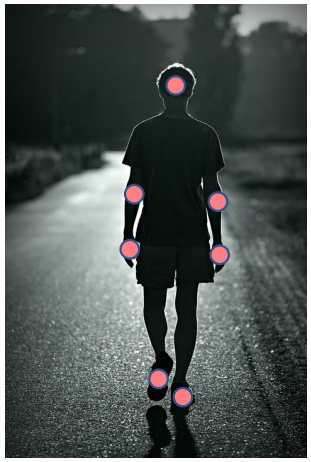
Classification

dog!

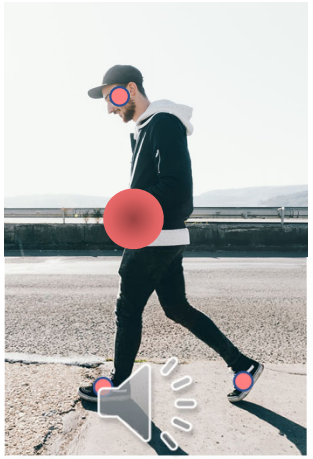


either dog or bear!

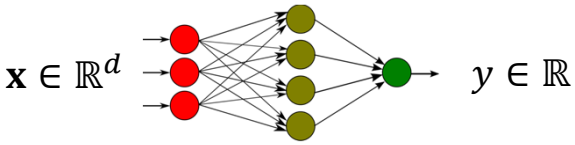




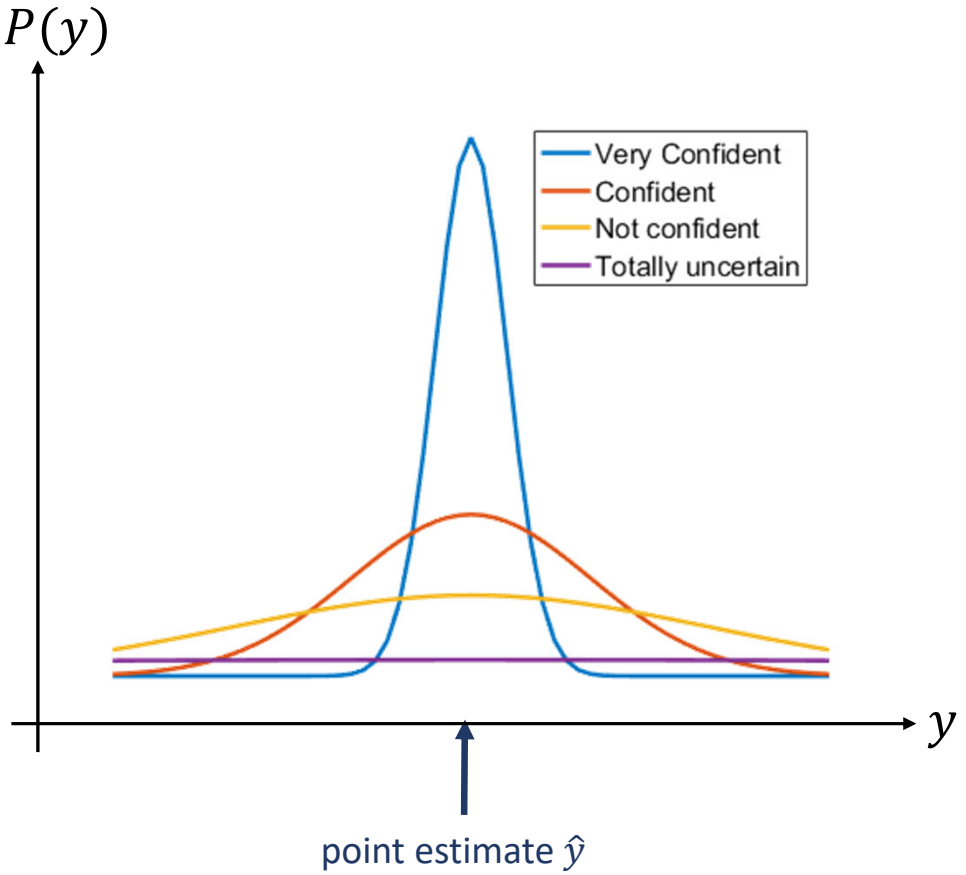
Regression



What do we mean by uncertainty estimation?



$f_{\boldsymbol{\theta}}(\mathbf{x}) \in \mathbb{R}^d \times \mathbb{R}$



Where can we use predictive uncertainty?



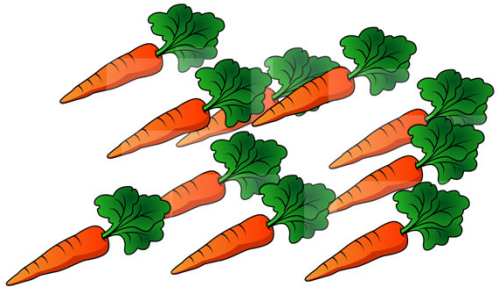
Where can we use predictive uncertainty?

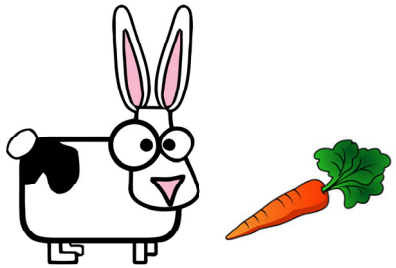
- Safety, reliability, transparency
  - Autonomous Driving
  - Medical Diagnosis
  - Drones
  - Judgement
  - High-frequency trading
  - Resource allocation (telecommunication, smart city, etc)
  - Defense against adversarial attacks
  - …



Where can we use predictive uncertainty?

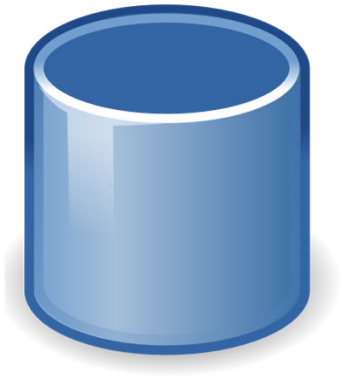
- Exploration vs Exploitation
  - Reinforcement Learning
  - Function approximation





Where can we use predictive uncertainty?

- Data Collection
  - Active Learning
  - Online continuous learning



Where can we use predictive uncertainty?

- Model diagnosis
  - Instead of analyzing the classification/regression error
    - Where is it wrong and highly confident?
    - Where it is right and under-confident?



Where can we use predictive uncertainty?

- Supervised Learning Tasks
  - multi-task learning
  - semi-supervised learning
  - transfer learning
  - anomaly detection
  - clustering
  - data compression
  - …

